

1 MinHash Signatures Generation

Given the 4 documents S_1, S_2, S_3, S_4 :

$$S_1 = \{a, b, e\}; \quad S_2 = \{a, b, d\}; \quad S_3 = \{c, d, e\}; \quad S_4 = \{c, e, f\},$$

- (A) Generate the bit-vector representations for each document (W5.1 - slide 21 & 22).
 (B) Generate the MinHash values of each document using the permutation $\pi = \{3, 0, 5, 2, 4, 1\}$ (W5.1 - Slide 23 to 30).

(1) universe set $\{a, b, c, d, e, f\}^{(2)}$

ID		S_1	S_2	S_3	S_4	π
0	a	1	1	0	0	3
1	b	1	1	0	0	0
2	c	0	0	1	1	5
3	d	0	1	1	0	2
4	e	1	0	1	1	4
5	f	0	0	0	1	1

2 Fast MinHash Signatures Generation

Consider the following three random universal hash functions for random permutations simulations. Please complete the following tasks.

$$h_1(x) = (4x + 1) \mod 6$$

$$S_1 = \{a, b, e\} \quad S_4 = \{c, e, f\}$$

$$h_2(x) = (5x + 1) \mod 6$$

$$S_2 = \{a, b, d\}$$

$$h_3(x) = (2x + 5) \mod 6$$

$$S_3 = \{c, d, e\}$$

- (A) Generate simulated permutations using three universal hash functions above.
 (B) Which of these hash functions gives the true permutations?
 (C) Generate the MinHash values using three universal hash functions above.
 (D) How effective is the signature representation preserving the all-pairs Jaccard similarity?

(A)

ID		S_1	S_2	S_3	S_4	$h_1(x)$	$h_2(x)$	$h_3(x)$
0	a	1	1	0	0	1	1	5
1	b	1	1	0	0	5	0	1
2	c	0	0	1	1	3	5	3
3	d	0	1	1	0	1	4	5
4	e	1	0	1	1	5	3	1
5	f	0	0	0	1	3	2	3

(B) $h_2(x)$ no repeat

(C)

ID		S_1	S_2	S_3	S_4	π_1	π_2	π_3
0	a	1	1	0	0	1	1	5
1	b	1	1	0	0	5	0	1
2	c	0	0	1	1	3	5	3
3	d	0	1	1	0	1	4	5
4	e	1	0	1	1	5	3	1
5	f	0	0	0	1	3	2	3

	S_1	S_2	S_3	S_4
$h_1(x)$	1	1	1	3
$h_2(x)$	0	0	3	2
$h_3(x)$	1	1	1	1

$$(D) I(S, T) = \frac{|S \cap T|}{|S \cup T|}$$

$$\text{true } J(S(S_1, S_2)) = \frac{2}{4} = \frac{1}{2}$$

① true J_S .

	S_1	S_2	S_3	S_4
S_1	1	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{3}$
S_2	..	1	$\frac{1}{2}$	0
S_3	.	..	1	$\frac{1}{2}$
S_4	.	..	.	1

3 Similarity / Distance Measures

Given two text documents A and B, which contain the following phrases:

- Document A: {"machine learning", "artificial intelligence", "data science", "deep learning"}
- Document B: {"artificial intelligence", "machine learning", "neural networks", "deep learning", "big data"}

(A) Calculate the Jaccard similarity between these two documents (W6 - Slide 5 to 7).

(B) Convert the two documents into vectors (consider each unique phrase as a dimension) and calculate their Cosine similarity. We can consider the order as {"machine learning", "artificial intelligence", "data science", "deep learning", "neural networks", "big data"} (W6 - Slide 9).

(C) Calculate the SimHash signatures for both documents using the document vectors from (B) and the random vectors provided below (W6 - Slide 12 & 13).

$$r_1 = [0.5, -0.1, 0.8, 0.2, -0.6, 0.4]$$

$$r_2 = [-0.3, 0.9, -0.2, 0.7, -0.8, 0.1]$$

$$r_3 = [0.7, 0.3, -0.4, 0.5, 0.2, -0.5]$$

Note: Cosine similarity is the cosine of the angle between the vectors; that is, it is the dot product of the vectors divided by the product of their lengths¹.

$$A) = \frac{3}{6} = \frac{1}{2}$$

universe set { machine, ar, data, deep, ne, bi }

$$(B) \quad x(1, 1, 1, 1, 0, 0)$$

$$y(1, 1, 0, 1, 1, 1)$$

$$\downarrow$$

$$x'(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$$

$$y'(\frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}})$$

$$\cos \theta = \sum_{i=1}^d x'_i y'_i = \frac{1}{2\sqrt{5}} + \frac{1}{2\sqrt{5}} + \frac{1}{2\sqrt{5}} = \frac{3}{2\sqrt{5}} \checkmark$$

> 0

$\angle = 1$

Cosine Similarity

$$= \frac{A \cdot B}{|A| \cdot |B|}$$

$$= \frac{x \cdot y}{|x| |y|}$$

$$= \frac{3}{2\sqrt{5}}$$

$$\sqrt{x^2} = 1$$

$$\sqrt{y^2} = 1$$

¹https://en.wikipedia.org/wiki/Cosine_similarity

(C). Simhash. $\begin{cases} 1 & \cos \theta \geq 0 \\ 0 & \cos \theta < 0 \end{cases}$

$$pr(x=y) = 1 - \frac{\theta xy}{\pi}$$

只确定正负

$$(A) r_1: 1 \times 0.5 - 1 \times 0.1 + 1 \times 0.8 + 1 \times 0.2 + 0 + 0$$

分母 > 0 只看分子 $r_2, r_3 = 1.4 > 0 = 1$

4 Parameters Tuning for Effective LSH

Consider a $(d_1, d_2, 1 - (1 - s_1^r)^b, 1 - (1 - s_2^r)^b)$ -sensitive family. Let the distance thresholds $d_1 = 0.7$ and $d_2 = 0.3$. The distance value is one minus the similarity. Below shows the collision probabilities for different similarity values s in the range of $[0,1]$ for different settings of (r,b) .

s	$(r = 2, b = 10)$	$(r = 3, b = 10)$	$(r = 5, b = 30)$
0.0	0	0	0
0.1	0.095617925	0.0099551198	2.999565×10^{-4}
0.2	0.33516736	0.077180588	0.009555888
0.3	0.61058388	0.23944889	0.070388681
0.4	0.82509877	0.48387073	0.2656604
0.5	0.94368649	0.73692442	0.61420956
0.6	0.98847078	0.91226748	0.91183045
0.7	0.99880958	0.98501511	0.99599499
0.8	0.99996344	0.99923405	0.99999328
0.9	0.99999994	0.99999786	1
1	1	1	1

- (A) Which setting gives the false negatives of 70%-similar pairs at most 5%?

$1 - p(s)$
- (B) Which setting gives the false positives of 30%-similar pairs at most 15%?

$p(s)$

Note: The LSH is $(d_1, d_2, 1 - (1 - s_1^r)^b, 1 - (1 - s_2^r)^b)$ -sensitive, where d_1 and d_2 correspond to the 70% and 30% similarities, respectively. i.e., there are two thresholds – the pairs with similarity $\geq 70\%$ are considered truly similar, and the pairs with similarity $\leq 30\%$ are considered truly dissimilar.

(A).

$1 - 0.998 = 0.2\% \quad \checkmark$
 $1 - 0.985 \approx 1.5\% \quad \checkmark$
 $1 - 0.995 \approx 0.5\% \quad \checkmark$

(B).

$0.71 > 15\%$
 $0.23 > 15\%$
 $0.07 < 15\% \quad \checkmark$